G.C.E. (O/L) · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 01

Mathematics II

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Time: Three hours (plus 10 minutes additional reading time) **Total marks:** 100

Index number: .....

Important:

- Use the additional 10 minutes to read the paper, select the questions and decide the order in which you will answer them.
- Select **five** questions from Part A and **five** questions from Part B, and answer **ten** questions in all.
- Write your answers on separate answer sheets. Where a question says “copy the figure”, copy it onto your answer sheet.
- Show the relevant steps and write the correct units when answering.
- Each question carries 10 marks. The marks for each part are given at its right-hand side.
- Mathematical tables, a straight edge with a cm/mm scale and a pair of compasses may be used. Graph paper will be provided.

Formulae: *The volume of a right pyramid of base area A and height h is $\frac{1}{3}Ah$.*

The volume of a right circular cylinder of base radius r and height h is $\pi r^2 h$.

Part A

Answer only five questions.

1. Kasun wants to buy a freezer for his shop. The cash price of the freezer is Rs 48 000. Two ways of paying for it are described in the table below.

Method 1 – Bank loan	Method 2 – Instalments
Borrow Rs 48 000 from a bank for one year at an annual simple interest rate of 10%. The loan and the interest are repaid in one payment at the end of the year.	Pay Rs 51 000 + Rs 120 = Rs 51 120 in 12 equal monthly instalments, that is the price plus Rs 3 120 interest. Each instalment repays Rs 4 000 of the price and a part of the interest. The interest is calculated on the unpaid balance (reducing balance) each month.

(i) Find the interest and the total amount paid under Method 1. (Marks 2)

(ii) Find the amount of each monthly instalment under Method 2. (Marks 1)

(iii) The unpaid balance of the price at the beginning of the 1st month is Rs 48 000, at the beginning of the 2nd month Rs 44 000, and so on. Show that the sum of the 12 monthly unpaid balances is Rs 312 000. (Marks 3)

(iv) The interest under Method 2 equals (sum of the monthly balances) $\times \frac{r}{100} \times \frac{1}{12}$. Using this, find the annual interest rate under Method 2, and show that it is higher than the rate under Method 1. (Marks 4)

2. An incomplete table of values for the function $y = f(x)$ is given below.

x	−5	−4	−3	−2	−1	0	1	2	3
y	−7	0	5	8	9	…	5	0	−7

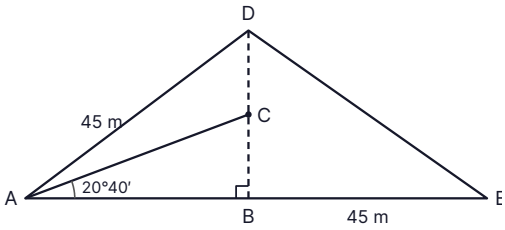
(i) Using the symmetry of the graph, find the value of y when $x = 0$. (Marks 1)

(ii) Choose a suitable scale and draw the graph of the function for $-5 \le x \le 3$ on graph paper. (Marks 3)

(iii) (a) Write the x-coordinates of the points at which the graph cuts the x-axis. (Marks 2)
(b) Using them, write the function in the form $y = -(x + p)(x + q)$.

(iv) The straight line $y = 6$ cuts the graph at the points A and B. (Marks 4)
(a) Draw the line $y = 6$ and write the range of values of x for which $y > 6$.
(b) Find the length of AB to the nearest first decimal place.

3. The figure shows a vertical cross-section of a hillside on Ravi's farm. A, B and E are on level ground. BD is vertical and C is a point on BD such that DC = BC. AC = BE = 45 m and CÂB = 20°40′. (Copy the figure on your answer sheet.) Use the mathematical tables to show that the angle of inclination of DE to the horizontal is greater than the angle of inclination of AC by more than 14°.



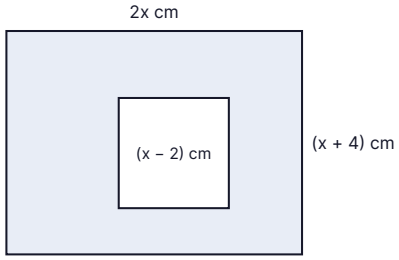
(i) Find the length of BC, and hence of BD. (Marks 3)

(ii) Write the value of tan DÊB. (Marks 2)

(iii) Using the tables, find DÊB to the nearest minute. (Marks 3)

(iv) Hence show that DÊB − CÂB > 14°. (Marks 2)

4. A rectangular signboard of length 2x cm and width (x + 4) cm has a square hole of side (x − 2) cm cut out of it, as shown in the figure. The area of the board that remains is 152 cm^2.



(i) Show that x satisfies the equation $x^2 + 12x - 156 = 0$. (Marks 3)

(ii) Solve the equation and find the value of x to the nearest cm. (Take $\sqrt{3} = 1.73$.) (Marks 4)

(iii) Using the positive root of the equation, show that the length 2x of the board is more than 15 cm. (Marks 3)

5. Nadeesha bought some small notebooks at Rs 90 each and some large notebooks at Rs 140 each for her class. She bought 30 notebooks altogether, and paid Rs 3 250 for them.

(i) Taking the number of small notebooks as x and the number of large notebooks as y, form a pair of simultaneous equations, and solve them to find the number of notebooks of each kind. (Marks 5)

(ii) Every small notebook has the same number of pages, and every large notebook has 40 pages more than a small notebook. The total number of pages in all the notebooks is 2 840. Find the number of pages in a small notebook. (Marks 5)

6. The table shows the number of customers who visited Dilini's shop on each of 40 days.

Customers per day	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79
Number of days	2	8	10	10	8	2

(i) Find the mean number of customers per day. (Use the mid-values of the classes.) (Marks 4)

(ii) Using the mean, find the expected total number of customers in 30 days. (Marks 2)

(iii) Each customer spends Rs 120 on average. Find the expected income of the shop in 7 days. (Marks 2)

(iv) Find the minimum possible total number of customers who came on the days when fewer than 40 customers came. (Marks 2)

Part B

Answer only five questions.

7. Use only a straight edge with a cm/mm scale and a pair of compasses for the following constructions. Construction lines should be clearly visible.

(i) Construct the line segment AB = 7 cm. At A, construct the ray AP such that ∠BAP = 90°. (Marks 2)

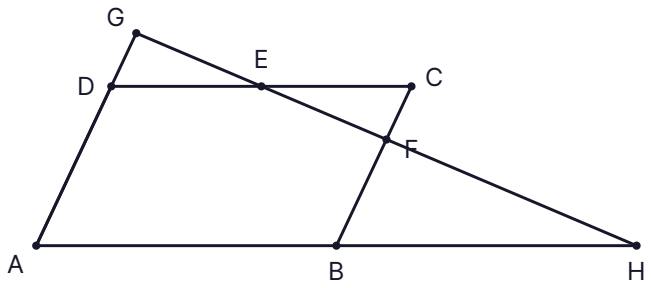
(ii) Bisect ∠BAP. Mark the point C on the bisector such that AC = 6 cm, and join BC. (Marks 2)

(iii) Construct the perpendicular bisector of BC, and with BC as a diameter draw a circle. (Marks 3)

(iv) Name as D the point, other than B, at which the circle meets AB. Join CD, and write the size of ∠BDC with the reason. (Marks 2)

(v) Find the size of ∠ACD. (Marks 1)

8. In the parallelogram ABCD, E is the midpoint of DC and F is the point on BC such that CF = $\frac{1}{3}$ BC. AD produced and FE produced meet at G, and AB produced and FE produced meet at H. (Copy the figure on your answer sheet.)



(i) Show that the triangles GDE and FCE are congruent. (Marks 3)

(ii) Join DF and GC, and show that GDCF is a parallelogram. (Hint: show that its diagonals bisect each other.) (Marks 2)

(iii) Show that the triangles HBF and GDE are equiangular. (Marks 3)

(iv) Show that BH = AB. (Marks 2)

9. Eight identical solid metal pyramids, each with a square base of side a cm and a height of $3a$ cm, are melted without any loss and recast into a solid right circular cylinder of base radius r cm and height $6r$ cm.

(i) Show that $a^3 = \frac{3}{4}\pi r^3$. (Marks 4)

(ii) Taking $r = 4.5$ and $\pi = 3.14$, find the value of a^3 using logarithms, and hence find a to the nearest cm. (Marks 3)

(iii) A solid cuboid has a square base of side a cm and a height of $3a$ cm. A pyramid with the same square base and a height of $3a$ cm is cut out of it. Using the value of a you found, find the volume of the metal that remains. (Marks 3)

10. Kasun and Dilini collect plastic bottles for a school recycling project. Kasun collected 8 bottles on the first day, and on each following day he collected 4 bottles more than on the previous day. He finished his collection on the 20th day.

(i) Write the number of bottles he collected on the 1st, 2nd and 3rd days. (Marks 1)

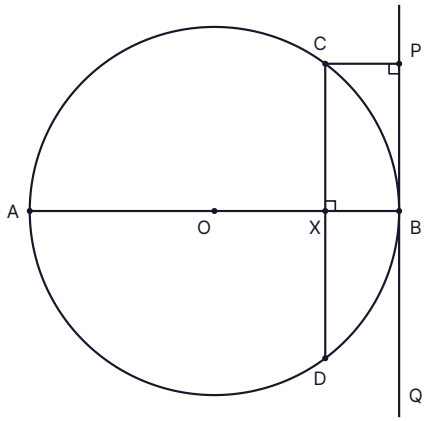
(ii) Find the number of bottles he collected on the 20th day. (Marks 2)

(iii) Find the total number of bottles he collected. (Marks 2)

(iv) Dilini collected the same total number of bottles in 16 days. On each day she collected 7 bottles more than on the previous day. Find the number of bottles she collected on the first day. (Marks 3)

(v) On which day did both of them collect the same number of bottles? (Marks 2)

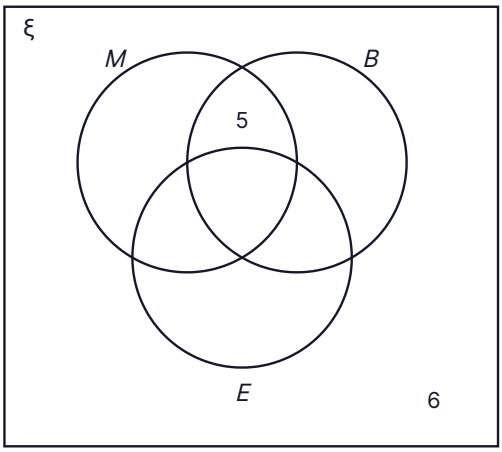
11. In the figure, O is the centre of the circle and AB is a diameter. The chord CD meets AB at X at right angles. PBQ is the tangent to the circle at B, and CP is perpendicular to PBQ. (Copy the figure on your answer sheet and join OC.)



(i) (a) Show that PBXC is a cyclic quadrilateral. (Marks 6)
(b) Show that BC bisects ∠PCO.

(ii) Name two angles that are equal to ∠DBQ, giving the reason for each. (Marks 4)

12. In a survey of 80 customers who visited Ravi's shop, 6 customers bought none of milk powder (M), bread (B) and eggs (E). 28 customers bought at least two of the three items, and 38 customers bought eggs. 5 customers bought only milk powder and bread. (The Venn diagram shows this information; copy it on your answer sheet.)



(i) Find the number of customers who bought exactly one of the three items. (Marks 3)

(ii) The number of customers who bought all three items is equal to the number who bought exactly two items. Find the number who bought all three. (Marks 3)

(iii) The number who bought only bread and eggs is twice the number who bought only milk powder and eggs. Find these two numbers. (Marks 2)

(iv) 34 customers bought milk powder. Find the number of customers who bought only bread. (Marks 2)

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